3.31 Formal Power Series and Laurent Series

Definition 3.31.1 (Formal Laurent Series)
*Let A be a commutative ring. Define the **formal Laurent series ring** $A((X))$ to be the set of sequences $(c_n)_{n\in\mathbb{Z}}$ in A such that there exists N for which $n \le N \implies c_n = 0$. Addition is defined pointwise with product defined as follows*

$$\left(\sum_{n\in\mathbb{Z}} c_n X^n\right)\left(\sum_{n\in\mathbb{Z}} d_n X^n\right) = \sum_{n\in\mathbb{Z}}\left(\sum_{k\in\mathbb{Z}} c_{n-k}d_k\right)X^n$$

where the inner sum is a sum of a family with all but finitely many terms zero. That this is well-defined and makes $A((X))$ into a commutative ring is shown in (3.31.3).

Definition 3.31.2 (Order of a Formal Laurent Series)
*Let A be a commutative ring. For $0 \neq h = \sum_n c_n X^n \in A((X))$ define the **order** $v(h) := \min\{n \mid c_n \neq 0\}$, and $v(0) := \infty$. The minimum exists since the c_n vanish for n sufficiently negative and $h \neq 0$. We call $c_{v(h)}$ the **leading coefficient** of h.*

Proposition 3.31.3 ($A((X))$ is a Commutative Ring)
Let A be a commutative ring and $c, d \in A((X))$. For every $n \in \mathbb{Z}$ the family $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, so $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined, and $(cd)_n = 0$ for $n < v(c) + v(d)$. In particular $cd \in A((X))$. Moreover $A((X))$ is a commutative ring under pointwise addition and this product.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The product is well-defined. Let $c = (c_n)_{n\in\mathbb{Z}}$ and $d = (d_n)_{n\in\mathbb{Z}}$ be elements of $A((X))$. If $c = 0$ or $d = 0$ every term below is zero and the product is 0, so assume $v(c), v(d)$ are finite. Fix $n \in \mathbb{Z}$. If the term $c_{n-k}d_k$ is nonzero then $n - k \ge v(c)$ and $k \ge v(d)$, that is

$$v(d) \le k \le n - v(c)$$

so only finitely many $k \in \mathbb{Z}$ give a nonzero term. Hence $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, and the inner sum $e_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined by (3.2.48). Moreover if $n < v(c) + v(d)$ then there is no integer k with $v(d) \le k \le n - v(c)$, so every term vanishes and $e_n = 0$. Therefore $(e_n)_{n\in\mathbb{Z}}$ is a sequence in A vanishing for $n < v(c) + v(d)$, that is an element of $A((X))$.

Ring axioms. Write $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ for the coefficients of the product, an element of $A((X))$ by the first part.

Addition. For $c, d \in A((X))$ we have $c_n + d_n = 0$ and $-c_n = 0$ for $n < \min(v(c), v(d))$, so $c + d$ and $-c$ lie in $A((X))$, as does the zero sequence. So $A((X))$ is a subgroup of the abelian group $A^{\mathbb{Z}}$ under pointwise addition.

Identity. Let $1 := (\delta_{n,0})_{n\in\mathbb{Z}}$, which vanishes for $n \le -1$ and so lies in $A((X))$. For $c \in A((X))$ the family $(\delta_{n-k,0}c_k)_{k\in\mathbb{Z}}$ has $k = n$ as its only possibly nonzero term, so $(1c)_n = c_n$.

Commutativity. The map $k \mapsto n - k$ is a bijection of $\mathbb{Z}$, so by (3.2.49) and commutativity of A,

$$(cd)_n = \sum_{k\in\mathbb{Z}} c_{n-k}d_k = \sum_{k\in\mathbb{Z}} c_k d_{n-k} = \sum_{k\in\mathbb{Z}} d_{n-k}c_k = (dc)_n$$

Distributivity. For $c, d, d' \in A((X))$, the families $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ and $(c_{n-k}d'_k)_{k\in\mathbb{Z}}$ have all but finitely many terms zero, so by (3.2.49)

$$(c(d+d'))_n = \sum_{k\in\mathbb{Z}} c_{n-k}(d_k + d'_k) = \sum_{k\in\mathbb{Z}} c_{n-k}d_k + \sum_{k\in\mathbb{Z}} c_{n-k}d'_k = (cd)_n + (cd')_n$$

and the other distributive law follows by commutativity.

Associativity. Let $c, d, e \in A((X))$ and fix $n \in \mathbb{Z}$; if any of them is 0 both sides are 0, so assume $v(c), v(d), v(e)$ are finite. Set $g(j,k) := c_{n-j-k}d_j e_k$ for $(j,k) \in \mathbb{Z}\times\mathbb{Z}$. If $g(j,k) \neq 0$ then $j \ge v(d)$, $k \ge v(e)$ and $n - j - k \ge v(c)$, so $v(d) \le j \le n - v(c) - v(e)$ and $v(e) \le k \le n - v(c) - v(d)$; hence g has all but finitely many terms zero. Recall that $(cd)_m = \sum_{j\in\mathbb{Z}} c_{m-j}d_j$ and $(de)_m = \sum_{k\in\mathbb{Z}} d_{m-k}e_k$. Then

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$$\begin{aligned}((cd)e)_n &= \sum_{k\in\mathbb{Z}} (cd)_{n-k}e_k && \text{by definition of the product}\\ &= \sum_{k\in\mathbb{Z}}\left(\sum_{j\in\mathbb{Z}} c_{n-k-j}d_j\right)e_k && \text{by definition of } (cd)_{n-k}\\ &= \sum_{k\in\mathbb{Z}}\sum_{j\in\mathbb{Z}} g(j,k) && \text{by (3.3.17)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.49)}\end{aligned}$$

and on the other side

$$\begin{aligned}(c(de))_n &= \sum_{l\in\mathbb{Z}} c_{n-l}(de)_l && \text{by definition of the product}\\ &= \sum_{l\in\mathbb{Z}} c_{n-l}\sum_{k\in\mathbb{Z}} d_{l-k}e_k && \text{by definition of } (de)_l\\ &= \sum_{l\in\mathbb{Z}}\sum_{k\in\mathbb{Z}} g(l-k,k) && \text{by (3.3.17)}\\ &= \sum_{(l,k)\in\mathbb{Z}\times\mathbb{Z}} g(l-k,k) && \text{by (3.2.49)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.49) for the bijection } (l,k)\mapsto(l-k,k)\end{aligned}$$

Hence $((cd)e)_n = (c(de))_n$ for all n. ∎

Lemma 3.31.4 (Valuation of a Sum and a Product)
Let A be a commutative ring and $c, d \in A((X))$. Then $v(1) = 0$, $v(-c) = v(c)$, and

a) $v(c+d) \geq \min(v(c), v(d))$
b) $v(cd) \geq v(c) + v(d)$
c) if A is an integral domain then so is $A((X))$, and $v(cd) = v(c) + v(d)$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Clearly $v(1) = 0$ and $v(-c) = v(c)$. If $n < \min(v(c), v(d))$ then $c_n = d_n = 0$, so $(c+d)_n = 0$, which gives a). For b), if $c = 0$ or $d = 0$ then $cd = 0$ and both sides are ∞. Otherwise $(cd)_n = 0$ for $n < v(c) + v(d)$ by (3.31.3), which means $v(cd) \geq v(c) + v(d)$. For c), let $c, d \neq 0$ with $m := v(c)$ and $r := v(d)$. A term $c_{m+r-k}d_k$ of $(cd)_{m+r}$ is nonzero only if $k \geq r$ and $m + r - k \geq m$, that is $k = r$, so $(cd)_{m+r} = c_m d_r \neq 0$ as A is a domain, while $(cd)_n = 0$ for $n < m + r$. Hence $cd \neq 0$ and $v(cd) = m + r$. ∎

Definition 3.31.5 (Formal Power Series)
*Let A be a commutative ring. Define the **formal power series ring** $A[[X]]$ to be the subring of $A((X))$ consisting of those sequences $(c_n)_{n\in\mathbb{Z}}$ for which $n < 0 \implies c_n = 0$, or equivalently those $h \in A((X))$ for which $v(h) \geq 0$.*

Proof that $A[[X]]$ is a subring of $A((X))$. Drafted by Claude Sonnet, reviewed by David Rufino.

$A[[X]] = \{h \in A((X)) \mid v(h) \geq 0\}$, so by (3.31.4) it contains 1 and is closed under negation, addition and multiplication. ∎

Lemma 3.31.6 (Formal Power Series Rings are Separated)
Let A be a commutative ring. For $N \geq 0$ we have $X^N A[[X]] = \{h \in A[[X]] \mid v(h) \geq N\}$, and $\bigcap_{N\geq 0} X^N A[[X]] = 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

If $h = X^N h'$ with $h' \in A[[X]]$ then the coefficients of h below X^N vanish, so $v(h) \geq N$. Conversely if $v(h) \geq N$ then $h = X^N h'$ where h' is h with the coefficients shifted down by N, an element of $A[[X]]$. If h lies in every $X^N A[[X]]$ then $v(h) \geq N$ for every N, so every coefficient of h vanishes and $h = 0$. ∎

Definition 3.31.7 (Formal Derivative)
Let A be a commutative ring. For $f = \sum_n c_n X^n \in A((X))$, define $f' := \sum_n n c_n X^{n-1} \in A((X))$, that is $(f')_n = (n+1)c_{n+1}$.

Lemma 3.31.8 (Valuation of a Derivative)
Let A be a commutative ring and $f \in A((X))$. Then $v(f') \geq v(f) - 1$. In particular if $f \in A[[X]]$ then $f' \in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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If $n < v(f) - 1$ then $n + 1 < v(f)$, so $c_{n+1} = 0$ and $(f')_n = (n+1)c_{n+1} = 0$. This gives $v(f') \geq v(f) - 1$. If $f \in A[[X]]$ then $v(f) \geq 0$ and $(f')_n = 0$ for $n < -1$; also $(f')_{-1} = 0 \cdot c_0 = 0$, so $f' \in A[[X]]$. ∎

Definition 3.31.9 (Summable Family in $A((X))$)
*Let A be a commutative ring. A family $(h_i)_{i\in I}$ in $A((X))$ is **summable** if for every $N \in \mathbb{Z}$ all but finitely many $i \in I$ satisfy $v(h_i) \geq N$. Its **sum** is the formal Laurent series*

$$\sum_{i\in I} h_i := \sum_{n\in\mathbb{Z}} \left(\sum_{i\in I} (h_i)_n \right) X^n$$

where $(h_i)_n$ denotes the coefficient of X^n in h_i.

Proof that the sum is well-defined. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and fix $n \in \mathbb{Z}$. By hypothesis (with $N = n+1$) the set $\{i \in I \mid v(h_i) \leq n\}$ is finite, and it contains every i with $h_{i,n} \neq 0$. So $(h_{i,n})_{i\in I}$ has all but finitely many terms zero and $s_n := \sum_{i\in I} h_{i,n}$ is defined by (3.2.48). Let $J_0 := \{i \in I \mid v(h_i) \leq 0\}$, which is finite, and $N_0 := \min\big(0, \min_{i\in J_0} v(h_i)\big)$ (with $N_0 := 0$ if J_0 is empty). If $n < N_0$ then $v(h_i) > n$ for every i, so every $h_{i,n} = 0$ and $s_n = 0$. Hence $(s_n)_{n\in\mathbb{Z}}$ lies in $A((X))$. ∎

Proposition 3.31.10 (Sums of Summable Families)
Let A be a commutative ring and $(h_i)_{i\in I}$ a summable family in $A((X))$. Then

a) If $v(h_i) \geq N$ for every $i \in I$ then $v\big(\sum_{i\in I} h_i\big) \geq N$. In particular if every $h_i \in A[[X]]$ then $\sum_{i\in I} h_i \in A[[X]]$.
b) For a bijection $\sigma : I' \to I$ the family $\big(h_{\sigma(i')}\big)_{i'\in I'}$ is summable and $\sum_{i'\in I'} h_{\sigma(i')} = \sum_{i\in I} h_i$.
c) The family $(h_i')_{i\in I}$ is summable and $\big(\sum_{i\in I} h_i\big)' = \sum_{i\in I} h_i'$.
d) For any $g \in A((X))$ the family $(gh_i)_{i\in I}$ is summable and $g \cdot \sum_{i\in I} h_i = \sum_{i\in I} gh_i$.
e) If $(k_i)_{i\in I}$ is another summable family then $(h_i + k_i)_{i\in I}$ is summable and $\sum_{i\in I}(h_i + k_i) = \sum_{i\in I} h_i + \sum_{i\in I} k_i$.
f) If $I = \bigsqcup_{\lambda\in\Lambda} I_\lambda$ is a partition of I into disjoint subsets then each $(h_i)_{i\in I_\lambda}$ and $\big(\sum_{i\in I_\lambda} h_i\big)_{\lambda\in\Lambda}$ is summable, and $\sum_{i\in I} h_i = \sum_{\lambda\in\Lambda} \sum_{i\in I_\lambda} h_i$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and $s_n := \sum_{i\in I} h_{i,n} = \big(\sum_{i\in I} h_i\big)_n$.

a) If $n < N$ then $h_{i,n} = 0$ for every i, so $s_n = 0$, as the sum of the zero family is 0 (3.2.48). Hence $v\big(\sum_{i\in I} h_i\big) \geq N$, and taking $N = 0$ gives the statement for $A[[X]]$.

b) The reindexed family clearly meets the summability condition. The family $(h_i)_{i\in I}$ itself is not almost zero in $A((X))$, but for each n the coefficient family $(h_{i,n})_{i\in I}$ in A is, so the statement holds coefficientwise by (3.2.49).

c) Since $v(h_i') \geq v(h_i) - 1$ by (3.31.8) the family $(h_i')_{i\in I}$ is also summable, so $\sum_{i\in I} h_i'$ is defined. Using $(h')_n = (n+1)h_{n+1}$, for every $n \in \mathbb{Z}$

$$\begin{aligned}
\left(\left(\sum_{i\in I} h_i \right)' \right)_n &= (n+1)s_{n+1} = (n+1) \sum_{i\in I} h_{i,n+1} && \text{by definition of the sum} \\
&= \sum_{i\in I} (n+1) h_{i,n+1} && \text{by (3.3.17)} \\
&= \sum_{i\in I} (h_i')_n = \left(\sum_{i\in I} h_i' \right)_n && \text{by definition of the sum}
\end{aligned}$$

d) Let $g \in A((X))$. If $g = 0$ both sides are 0, so assume $v(g)$ is finite. By (3.31.4) $v(gh_i) \geq v(g) + v(h_i)$, so $(gh_i)_{i\in I}$ is summable and $\sum_{i\in I} gh_i$ is defined. Fix $n \in \mathbb{Z}$ and set $a(k,i) := g_{n-k} h_{i,k}$ for $(k,i) \in \mathbb{Z} \times I$. If $a(k,i) \neq 0$ then $k \leq n - v(g)$ and $v(h_i) \leq k$, so i lies in the finite set $\{i \in I \mid v(h_i) \leq n - v(g)\}$ and $v(h_i) \leq k \leq n - v(g)$; hence a has all but finitely many terms zero. Then

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$$\begin{aligned}
\left(g\sum_{i\in I}h_i\right)_n &= \sum_{k\in\mathbb{Z}} g_{n-k}s_k = \sum_{k\in\mathbb{Z}} g_{n-k}\sum_{i\in I}h_{i,k} && \text{by definition of the product and the sum}\\
&= \sum_{k\in\mathbb{Z}}\sum_{i\in I} a(k,i) && \text{by (3.3.17)}\\
&= \sum_{i\in I}\sum_{k\in\mathbb{Z}} a(k,i) && \text{by (3.2.49)}\\
&= \sum_{i\in I}\sum_{k\in\mathbb{Z}} g_{n-k}h_{i,k} && \text{by definition of } a\\
&= \sum_{i\in I}(gh_i)_n = \left(\sum_{i\in I} gh_i\right)_n && \text{by definition of the product and the sum}
\end{aligned}$$

e) By (3.31.4) a), $v(h_i+k_i)\geq\min(v(h_i),v(k_i))$, so for every N all but finitely many i satisfy $v(h_i+k_i)\geq N$ and $(h_i+k_i)_{i\in I}$ is summable. The identity holds in each coefficient by (3.2.49).

f) The subfamilies are summable. For $N\in\mathbb{Z}$ the set $F_N := \{i\in I \mid v(h_i)<N\}$ is finite, so only finitely many λ have $I_\lambda\cap F_N\neq\emptyset$, and for every other λ we have $v\left(\sum_{i\in I_\lambda}h_i\right)\geq N$ by a). Hence $\left(\sum_{i\in I_\lambda}h_i\right)_{\lambda\in\Lambda}$ is summable. The identity holds in each coefficient by (3.2.49) applied to $(h_{i,n})_{i\in I}$. ∎

Corollary 3.31.11 (Interchange of Summation)
Let A be a commutative ring and $(a(i,j))_{(i,j)\in I\times J}$ a summable family in $A((X))$. Then each family $(a(i,j))_{j\in J}$ and $(a(i,j))_{i\in I}$ is summable, the families $\left(\sum_{j\in J}a(i,j)\right)_{i\in I}$ and $\left(\sum_{i\in I}a(i,j)\right)_{j\in J}$ are summable, and

$$\sum_{(i,j)\in I\times J} a(i,j) = \sum_{i\in I}\sum_{j\in J} a(i,j) = \sum_{j\in J}\sum_{i\in I} a(i,j).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Apply (3.31.10) f) to the partition of $I\times J$ into the rows $\{i\}\times J$, and then to the partition into the columns $I\times\{j\}$. ∎

Lemma 3.31.12 (Convergence Criterion for a Sum)
Let A be a commutative ring, $(h_j)_{j\geq 0}$ a summable family in $A((X))$ and $h\in A((X))$. Then $h=\sum_{j\geq 0}h_j$ if and only if the partial sums converge to h, that is $v\left(h-\sum_{j\leq J}h_j\right)\to\infty$ as $J\to\infty$ (for every $N\in\mathbb{Z}$ we have $v\left(h-\sum_{j\leq J}h_j\right)\geq N$ for all sufficiently large J).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $S:=\sum_{j\geq 0}h_j$ and fix $N\in\mathbb{Z}$. As the family is summable, there is J_0 with $v(h_j)\geq N$ for all $j>J_0$. For every J, partitioning $\mathbb{N}$ into $\{0,\dots,J\}$ and $\{j>J\}$ gives $S-\sum_{j\leq J}h_j=\sum_{j>J}h_j$ by (3.31.10), and for $J\geq J_0$ this has $v\geq N$ by part a) of that proposition.

If $h=S$ then $h-\sum_{j\leq J}h_j=\sum_{j>J}h_j$ has $v\geq N$ for every $J\geq J_0$, so the partial sums converge to h. Conversely suppose the partial sums converge to h, and choose $J\geq J_0$ with $v\left(h-\sum_{j\leq J}h_j\right)\geq N$. Then by (3.31.4)

$$v(h-S)\geq\min\left(v\left(h-\sum_{j\leq J}h_j\right),v\left(\sum_{j>J}h_j\right)\right)\geq N.$$

This holds for every N, so every coefficient of $h-S$ vanishes. ∎

Proposition 3.31.13 (Units of a Formal Power Series Ring)
Let A be a commutative ring and $u\in A[[X]]$, with constant coefficient u_0 (the coefficient of X^0). Then u is a unit of $A[[X]]$ if and only if $u_0\in A^\star$. In particular, when $A=k$ is a field, $u\in k[[X]]^\star\iff u_0\neq 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose $u_0\in A^\star$ and set $w:=1-u_0^{-1}u\in A[[X]]$. Its constant coefficient is 0, that is $v(w)\geq 1$, so $v(w^i)\geq i$ for all $i\geq 0$ by (3.31.4). Hence $(w^i)_{i\geq 0}$ is summable (3.31.9) and $s:=\sum_{i\geq 0}w^i\in A[[X]]$ by (3.31.10). Then

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$$\begin{aligned} s &= 1 + \sum_{i\geq 1} w^i && \text{splitting off } i=0 \text{ by (3.31.10) } f) \\ &= 1 + \sum_{i\geq 0} w^{i+1} && \text{reindexing } i \mapsto i+1 \text{ by (3.31.10) } b) \\ &= 1 + ws && \text{distributivity, by (3.31.10) } d) \end{aligned}$$

so $(1-w)s = 1$. Since $u_0^{-1}u = 1-w$, this says $u\cdot(u_0^{-1}s) = 1$, so $u_0^{-1}s$ is an inverse of u.

Conversely, if u is a unit with inverse $t \in A[[X]]$ then $(ut)_0 = \sum_{k\in\mathbb{Z}} u_{-k}t_k = u_0t_0$, as $u_{-k} = 0$ for $k > 0$ and $t_k = 0$ for $k < 0$. So $u_0t_0 = 1$ and $u_0 \in A^\star$. When $A = k$ is a field, $u_0 \in k^\star$ is equivalent to $u_0 \neq 0$. ∎

Proposition 3.31.14 (Units of a Formal Laurent Series Ring)
Let A be an integral domain and $0 \neq h \in A((X))$ with $m := v(h)$. The following are equivalent

a) h is a unit of $A((X))$
b) $h = X^m u$ for a unit $u \in A[[X]]^\star$
c) the leading coefficient h_m of h is a unit of A

In particular when $A = k$ is a field, $k((X))$ is a field.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

c) $\implies$ b): $u := X^{-m}h \in A[[X]]$ has constant coefficient $u_0 = h_m \in A^\star$, so u is a unit of $A[[X]]$ by (3.31.13).

b) $\implies$ a): $X^{-m}u^{-1}$ is an inverse of $h = X^m u$.

a) $\implies$ c): let $hh' = 1$ in $A((X))$. Then $v(h) + v(h') = v(1) = 0$ by (3.31.4) c), so $v(h') = -m$, and the coefficient of X^0 in hh' is $h_m h'_{-m} = 1$, so $h_m \in A^\star$. ∎

Proposition 3.31.15 ($k((X))$ is a Field of Fractions for $k[[X]]$)
Let k be a field. Then every element of the field $k((X))$ is a/b for $a, b \in k[[X]]$, $b \neq 0$ – that is, $k((X))$ is a field of fractions for the subring $k[[X]] \subset k((X))$. Moreover, the order function $v : k((X))^\star \to \mathbb{Z}$ is a group homomorphism under multiplication: $v(fg) = v(f) + v(g)$, and $v(f^{-1}) = -v(f)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.31.14), as k is a field every $c \neq 0$ in $k((X))$ is $c = X^m u$ with $m = v(c)$ and $u \in k[[X]]^\star$, which is $uX^m \in k[[X]]$ if $m \geq 0$ and u/X^{-m} otherwise, in either case a ratio of two elements of $k[[X]]$.

By (3.31.4) c), as k is a domain, $v(fg) = v(f) + v(g)$ for $f, g \neq 0$. Applying this to f, f^{-1}: $0 = v(1) = v(ff^{-1}) = v(f) + v(f^{-1})$, giving $v(f^{-1}) = -v(f)$. ∎

Proposition 3.31.16 (Formal Power Series Ring is a DVR)
Let k be a field. Then $k[[X]]$ is an integral domain, not a field, in which every nonzero element a can be written uniquely as $a = X^{v(a)}u$ with $u \in k[[X]]^\star$ (equivalently, $u(0) \neq 0$ (3.31.13)).

Using the criteria (3.29.2)e) $k[[X]]$ is a discrete valuation ring with valuation precisely the order function v.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$k[[X]]$ is a domain, as a subring of the domain $k((X))$ (3.31.4).c), and not a field, as $X \neq 0$ has no inverse ($v(Xg) = 1 + v(g) \neq 0$ for every $g \in k[[X]]$ by the same result). Given $0 \neq a \in k[[X]]$, by (3.31.14) $a = X^{v(a)}u$ with $u \in k[[X]]^\star$, where $v(a) \geq 0$: this is exactly the uniformised-element criterion (3.29.2)e), with $\pi := X$. If also $a = X^n u'$ with $u' \in k[[X]]^\star$ then $v(u') = 0$ and $v(a) = n + v(u') = n$ by (3.31.4) c), so the exponent is $v(a)$ and then $u' = u$ as $k((X))$ is a domain. ∎

The next result generalises the Taylor and Laurent series expansion of a meromorphic function at a point. A function f holomorphic near $p \in \mathbb{C}$ is expanded in powers of the local parameter $z - p$ as $f = \sum_{i\geq 0} c_i(z-p)^i$ with $c_i \in \mathbb{C}$, the residue field of the local ring at p, and a meromorphic function is expanded in the same way with finitely many negative powers. For a DVR $(A, \mathfrak{m})$ with uniformiser π we expand elements $a \in A$ in powers of π in the same way: the coefficients $c_i(a)$, which we call the **digits** of a (as in the base-p expansion of an integer, successive digits are determined by working modulo successive powers of $\mathfrak{m}$), are elements of the residue field $k(\mathfrak{m})$, included in the quotients $A/\mathfrak{m}^{n+1}$ by the sections of (3.24.18). It also corresponds to the p-adic expansion $a = \sum_i c_i p^i$ of an integer, or more generally of a rational number $x = \sum_{i\geq v_p(x)} c_i p^i$ with finitely many negative powers (the analogue of the Laurent series), in the DVR $\mathbb{Z}_{(p)}$ with uniformiser p, residue field $\mathbb{F}_p$ and digits $c_i \in \{0, \ldots, p-1\}$. There the digits are only a choice of representatives of $\mathbb{F}_p$ in $\mathbb{Z}_{(p)}$, as $\mathbb{Z}_{(p)}$ contains no copy of $\mathbb{F}_p$, and the expansion is not compatible with

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addition and multiplication (carries); in the present equal-characteristic situation the sections s_n remove this choice and make the expansion a ring homomorphism. This gives the "Taylor series" $\sum_i c_i(a)X^i \in k(\mathfrak{m})[[X]]$ of a, and its extension to the fraction field is the "Laurent series" of an element of K. For $k(\mathfrak{m}) = k$ and the local ring of a point of a curve over an algebraically closed field k, this is the expansion described in the text of [Lan82] (Ch. I, §2).

Proposition 3.31.17 (Coefficient Expansion of a DVR over a Field)
Let $(A, \mathfrak{m})$ be a DVR with fraction field K and valuation v, and π a uniformiser of A. Let k be a field and suppose A is a k-algebra whose residue field $k(\mathfrak{m}) = A/\mathfrak{m}$ is separable algebraic over k (for example if k is perfect and $k(\mathfrak{m})/k$ is algebraic (3.17.106)).

Recall the k-algebra sections $s_n : k(\mathfrak{m}) \to A/\mathfrak{m}^{n+1}$ of the reduction maps $\pi_n : A/\mathfrak{m}^{n+1} \to k(\mathfrak{m})$ from (3.24.18): they are unique, with $\pi_n \circ s_n = 1_{k(\mathfrak{m})}$, and compatible, $\rho_{n,n-1} \circ s_n = s_{n-1}$. For $a \in A$ write $[a]_n$ for its class in $A/\mathfrak{m}^{n+1}$.

*Then every $a \in A$ has a unique sequence $(c_i(a))_{i \geq 0}$ in $k(\mathfrak{m})$, the **digits** of a relative to π, such that for every $n \geq 0$*

$$[a]_n = \sum_{i=0}^{n} s_n(c_i(a))\,[\pi^i]_n \quad \text{in } A/\mathfrak{m}^{n+1}.$$

The map

$$L_\pi : A \to k(\mathfrak{m})[[X]], \quad L_\pi(a) := \sum_{i \geq 0} c_i(a) X^i,$$

is an injective k-algebra homomorphism with $L_\pi(\pi) = X$. Moreover $v_X(L_\pi(a)) = v(a)$ for every $0 \neq a \in A$, and L_π extends uniquely to an injective ring homomorphism $L_\pi : K \to k(\mathfrak{m})((X))$ (still denoted L_π).

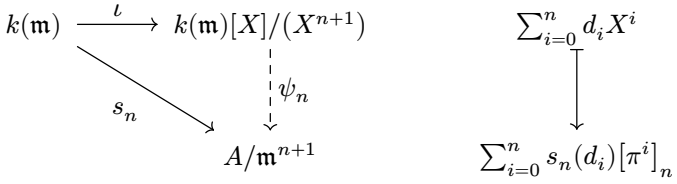
When $k(\mathfrak{m}) = k$, that is $k \to A/\mathfrak{m}$ is an isomorphism, s_n is induced by the structure map $k \to A$, so the displayed identity reads $[a]_n = \sum_{i=0}^{n} c_i [\pi^i]_n$ with $c_i \in k$, that is $a - \sum_{i=0}^{n} c_i \pi^i \in \mathfrak{m}^{n+1}$, and $L_\pi : A \to k[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Notation. We write $\overline{a} := [a]_0 \in k(\mathfrak{m})$ for the residue class of $a \in A$. In $k(\mathfrak{m})[X]/(X^{n+1})$ we write the class of X again as X.

The sections s_n. These exist, are unique and are compatible by (3.24.18). If $k(\mathfrak{m}) = k$ this is immediate: a k-algebra homomorphism out of k is unique, so s_n is the map induced by $k \to A$, which is a section of π_n, as $k \to A/\mathfrak{m}$ is an isomorphism.

The isomorphisms ψ_n. Fix $n \geq 0$. We claim that there is a unique ring homomorphism $\psi_n : k(\mathfrak{m})[X]/(X^{n+1}) \to A/\mathfrak{m}^{n+1}$ with $\psi_n(X) = [\pi]_n$ making the following diagram commute, where ι is the inclusion of the constants, and that ψ_n is an isomorphism:



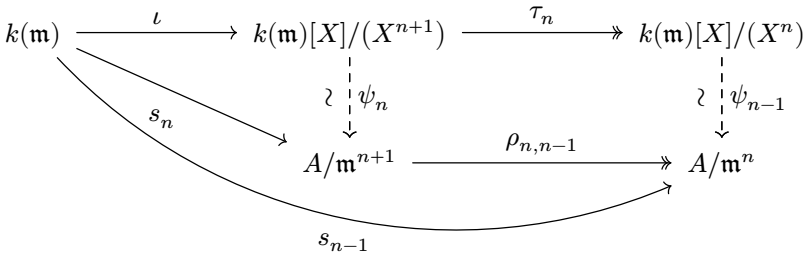
Existence and uniqueness. By the universal property of the polynomial ring $k(\mathfrak{m})[X]$ (3.7.6), applied to the ring homomorphism s_n and the element $[\pi]_n \in A/\mathfrak{m}^{n+1}$, there is a unique ring homomorphism $k(\mathfrak{m})[X] \to A/\mathfrak{m}^{n+1}$ restricting to s_n on $k(\mathfrak{m})$ with $X \mapsto [\pi]_n$, namely $\sum d_i X^i \mapsto \sum s_n(d_i)[\pi^i]_n$; it is a k-algebra homomorphism as s_n is. As $\pi^{n+1} \in \mathfrak{m}^{n+1}$ it kills X^{n+1}, so it factors uniquely through ψ_n. It remains to show that ψ_n is bijective.

Surjective. By induction on n, the case $n = 0$ being $\psi_0 = s_0 = (A/\mathfrak{m} = k(\mathfrak{m}))$. Given $[a]_n \in A/\mathfrak{m}^{n+1}$ with $a \in A$, its image in $A/\mathfrak{m}^n$ is $\psi_{n-1}(g)$ for some g; lifting g to $g' \in k(\mathfrak{m})[X]$ of degree $< n$, compatibility of the s_i gives $[a]_n - \psi_n(g') \in \mathfrak{m}^n/\mathfrak{m}^{n+1}$. Now $\mathfrak{m}^n = \pi^n A$, so this element is $[\pi^n a']_n$ for some $a' \in A$. Put $c := \overline{a'} \in k(\mathfrak{m})$. Then $[a']_n$ and $s_n(c)$ both reduce to c modulo $\mathfrak{m}$, so $[a']_n - s_n(c) \in \mathfrak{m}/\mathfrak{m}^{n+1}$, and multiplying by $[\pi^n]_n$, which lies in $\mathfrak{m}^n/\mathfrak{m}^{n+1}$, gives $[\pi^n a']_n = s_n(c)[\pi^n]_n$. Hence $[a]_n = \psi_n(g' + cX^n)$.

Injective. If $g = \sum_{i \leq n} d_i X^i$ is non-zero and d_j is its first non-zero coefficient, then $\psi_n(g) = [\pi^j]_n \big(s_n(d_j) + [\pi]_n \cdots\big)$ with $s_n(d_j)$ a unit of $A/\mathfrak{m}^{n+1}$ (it reduces to $d_j \neq 0$ in $k(\mathfrak{m})$), so a lift of $\psi_n(g)$ to A has valuation exactly $j \leq n$ and $\psi_n(g) \neq 0$.

Compatibility. Let $\tau_n : k(\mathfrak{m})[X]/(X^{n+1}) \to k(\mathfrak{m})[X]/(X^n)$ be the truncation, and ι the inclusions of the constants. We claim that the ψ_n are compatible with the reductions and with the s_n, that is the following diagram commutes:

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The triangle with s_n commutes by the definition of ψ_n, and the triangle with s_{n-1} by the definition of ψ_{n-1}, as $\tau_n \circ \iota$ is the inclusion of the constants; the remaining triangle $\rho_{n,n-1} \circ s_n = s_{n-1}$ is the compatibility of the sections. It remains to show that the square commutes. The homomorphism $\rho_{n,n-1} \circ \psi_n$ sends X^n to $\rho_{n,n-1}\big([\pi^n]_n\big) = [\pi^n]_{n-1} = 0$ in $A/\mathfrak{m}^n$, so it factors as $\chi \circ \tau_n$ for a ring homomorphism $\chi : k(\mathfrak{m})[X]/(X^n) \to A/\mathfrak{m}^n$. Then $\chi(X) = [\pi]_{n-1}$ and $\chi \circ \iota = \rho_{n,n-1} \circ \psi_n \circ \iota = \rho_{n,n-1} \circ s_n = s_{n-1}$, so χ makes the triangle for $n-1$ commute, and $\chi = \psi_{n-1}$ by the uniqueness in the claim for $n-1$.

The digits and L_π. Put $\phi_n := \psi_n^{-1}$. By the compatibility of the ψ_n, $\phi_{j-1} \circ \rho_{j,j-1} = \tau_j \circ \phi_j$, so $\phi_{j-1}\big([a]_{j-1}\big)$ is the truncation of $\phi_j\big([a]_j\big)$. Hence the coefficient of X^i in $\phi_j\big([a]_j\big)$ is the same for every $j \geq i$, namely $\phi_j\big([a]_j\big)_i = \phi_i([a]_i)_i$, and we define this to be the digit $c_i(a)$.

By construction $\phi_n([a]_n) = \sum_{i \leq n} c_i(a) X^i$ for every n, that is $[a]_n = \psi_n\Big(\sum_{i \leq n} c_i(a) X^i\Big)$, which is the displayed identity. The sequence is unique, as ψ_n is injective and so determines the coefficients $c_0, \ldots, c_n$ for each n.

For $a, b \in A$ and $i \leq n$, comparing coefficients of X^i in $\phi_n([a+b]_n) = \phi_n([a]_n) + \phi_n([b]_n)$ and $\phi_n([ab]_n) = \phi_n([a]_n)\phi_n([b]_n)$ gives $c_i(a+b) = c_i(a) + c_i(b)$ and $c_i(ab) = \sum_{j=0}^{i} c_j(a) c_{i-j}(b)$, and $c_i(1) = \delta_{i,0}$ similarly. So L_π is a ring homomorphism, and it is a k-algebra homomorphism as s_n is k-linear: the image of $c \in k$ is the constant series $\overline{c} \in k(\mathfrak{m})$. For π: $[\pi]_n = \psi_n(X)$, so $c_i(\pi) = \delta_{i,1}$ and $L_\pi(\pi) = X$ (for $n \geq 1$).

Valuation formula and injectivity. Let $0 \neq a \in A$, $m := v(a)$, so $a = u\pi^m$ for a unit $u \in A^\star$ (3.29.2), and write $\overline{u} \in k(\mathfrak{m})^\star$ for its residue. Then $[u]_m$ and $s_m(\overline{u})$ both reduce to $\overline{u}$ modulo $\mathfrak{m}$, so $[u]_m - s_m(\overline{u}) \in \mathfrak{m}/\mathfrak{m}^{m+1}$, and multiplying by $[\pi^m]_m$ gives $[a]_m = [u\pi^m]_m = s_m(\overline{u})[\pi^m]_m = \psi_m(\overline{u} X^m)$. So $\phi_m([a]_m) = \overline{u} X^m$, a polynomial with no terms below X^m. Hence $c_i(a) = 0$ for $i < m$ and $c_m(a) = \overline{u} \neq 0$, so $v_X(L_\pi(a)) = m = v(a)$. In particular $L_\pi(a) \neq 0$ for $a \neq 0$ and L_π is injective.

L_π extends to K. As $L_\pi : A \to k(\mathfrak{m})[[X]]$ is injective and $k(\mathfrak{m})[[X]] \subset k(\mathfrak{m})((X))$ (a field) (3.31.15), $A \setminus \{0\}$ maps into $k(\mathfrak{m})((X))^\star$. So (3.5.5) (with $S := A \setminus \{0\}$, so $S^{-1}A = K$) gives a unique ring homomorphism $L_\pi : K \to k(\mathfrak{m})((X))$ with $L_\pi|_A$ the original map, explicitly $L_\pi(a/s) = L_\pi(a) L_\pi(s)^{-1}$; it is injective (if $L_\pi(a/s) = 0$ then $L_\pi(a) = 0$, so $a = 0$, giving $a/s = 0$). ∎

Remark 3.31.18

The case $k(\mathfrak{m}) = k$ is the classical one, for instance $A = \mathcal{O}_{X,p}$ the local ring of a point of a curve over an algebraically closed field k (or any finitely generated k-algebra localised at a k-rational point, which is a DVR). Then $k \subseteq A$ is a coefficient field, the digits $c_i(a)$ lie in A itself, the displayed identity is an honest congruence $a \equiv \sum_{i \leq n} c_i \pi^i \pmod{\mathfrak{m}^{n+1}}$ in A, and $L_\pi(a) = \sum_i c_i(a) X^i \in k[[X]]$ is the Taylor series of the function a at p in the local parameter π, with Laurent series for elements of K. In general the digits lie in the residue field $k(\mathfrak{m})$, which need not embed in A: for $A = \mathbb{R}[t]_{(t^2+1)}$ the residue field is $\mathbb{C}$, but $A \subseteq \mathbb{R}(t)$ contains no copy of $\mathbb{C}$. The sections s_n lift digits only modulo $\mathfrak{m}^{n+1}$, the expansion lands in $k(\mathfrak{m})[[X]]$, and it is canonical only because the sections are unique, which is where separability of $k(\mathfrak{m})/k$ is used. This is what gives the residue symbol at a non-rational place of a curve over a perfect field.

Proposition 3.31.19 (Product Rule)

Let A be a commutative ring and $f, g \in A((X))$. Then $(fg)' = f'g + fg'$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $f = \sum_i a_i X^i$, $g = \sum_j b_j X^j$ and fix $n \in \mathbb{Z}$. All sums below are over $i \in \mathbb{Z}$ and are sums of families with all but finitely many terms zero, with the coefficients of a product as in (3.31.3). The coefficient of X^{n-1} in $(fg)'$ is $n(fg)_n = n \sum_i a_i b_{n-i}$. Since $(f')_k = (k+1) a_{k+1}$ and $(g')_k = (k+1) b_{k+1}$, reindexing $i = k+1$ and $i = n-k-1$ respectively (3.2.49) gives

$$(f'g)_{n-1} = \sum_i i a_i b_{n-i} \qquad (fg')_{n-1} = \sum_i (n-i) a_i b_{n-i}$$

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Hence

$$\begin{aligned}(f'g+fg')_{n-1} &= \sum_i i a_i b_{n-i} + \sum_i (n-i) a_i b_{n-i} && \text{by the above}\\ &= \sum_i n a_i b_{n-i} && \text{by (3.2.49)}\\ &= n\sum_i a_i b_{n-i} = ((fg)')_{n-1} && \text{by (3.3.17)}\end{aligned}$$

As n was arbitrary, $(fg)' = f'g + fg'$. ∎

Lemma 3.31.20 (Power Rule)
Let A be a commutative ring and $\psi \in A((X))$. Then $(\psi^n)' = n\psi^{n-1}\psi'$ for every $n \geq 0$. If moreover ψ is a unit of $A((X))$, this also holds for every $n < 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$n \geq 0$. Induction on n: the case $n = 0$ is $0 = 0$ ($\psi^0 = 1$ is constant). Given $(\psi^n)' = n\psi^{n-1}\psi'$, the product rule (3.31.19) gives

$$(\psi^{n+1})' = (\psi \cdot \psi^n)' = \psi'\psi^n + \psi(\psi^n)' = \psi'\psi^n + n\psi^n\psi' = (n+1)\psi^n\psi'$$

$n < 0$, ψ a unit. Differentiating $\psi\psi^{-1} = 1$ with the product rule, $0 = \psi'\psi^{-1} + \psi(\psi^{-1})'$, so $(\psi^{-1})' = -\psi^{-2}\psi'$ – the case $n = -1$. For $n \leq -2$, write $n = -r$, $r \geq 2$: applying the $n \geq 0$ case just shown to the element $\psi^{-1} \in A((X))$ (no hypothesis on its order was used there),

$$(\psi^{-r})' = \left((\psi^{-1})^r\right)' = r(\psi^{-1})^{r-1}(\psi^{-1})' = r(\psi^{-1})^{r-1}(-\psi^{-2}\psi') = -r\psi^{-r-1}\psi' = n\psi^{n-1}\psi'$$

∎

Proposition 3.31.21 (Substitution into a Formal Laurent Series)
Let A be a commutative ring, $f = \sum_{n \geq n_0} c_n X^n \in A((X))$ and $\psi \in XA[[X]]$ (that is, $\psi(0) = 0$). Suppose that either

- *$f \in A[[X]]$, that is $n_0 \geq 0$, or*
- *A is an integral domain and $\psi \in A((X))^\star$, that is $\psi \in X^m A[[X]]^\star$ for some $m > 0$ (3.31.14)*

Then

$$(f \circ \psi)(X) := \sum_{n \geq n_0} c_n \psi(X)^n$$

is a well-defined element of $A((X))$ (with ψ^n, $n < 0$, the power in the ring $A((X))$), and $f \circ \psi \in A[[X]]$ if $f \in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\psi \in XA[[X]]$ has no terms of degree ≤ 0 we have $v(\psi) \geq 1$, so $v(\psi^n) \geq n$ for every $n \geq 0$ by (3.31.4) b). Negative powers of ψ only occur for the finitely many n with $n_0 \leq n < 0$. Hence for every N all $n \geq \max(N, 0)$ satisfy $v(c_n\psi^n) \geq n \geq N$, and only the finitely many n with $n_0 \leq n < \max(N, 0)$ can fail this, so $(c_n\psi^n)_{n \geq n_0}$ is summable (3.31.9) and its sum lies in $A((X))$. If $f \in A[[X]]$ then every $c_n\psi^n \in A[[X]]$ and the sum lies in $A[[X]]$ by (3.31.10) a). ∎

Proposition 3.31.22 (Chain Rule)
In the setting of (3.31.21), $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By the power rule (3.31.20) (which for $n < 0$ requires that ψ is a unit of $A((X))$, as it is then), $(c_n\psi^n)' = c_n(\psi^n)' = nc_n\psi^{n-1}\psi'$ for every $n \geq n_0$, using linearity of $(-)'$ (immediate from (3.31.7), as $c_n \in A$ is a constant). The term $n = 0$ is 0, and is simply omitted if ψ is not a unit. Then, by termwise differentiation (3.31.10) c) of the summable family $(c_n\psi^n)_{n \geq n_0}$ from the proof of (3.31.21),

$$(f \circ \psi)' = \sum_{n \geq n_0} (c_n\psi^n)' = \sum_{n \geq n_0} nc_n\psi^{n-1}\psi'$$

The family $(nc_n\psi^{n-1})_{n \geq n_0}$ is summable by the same order bound as $(c_n\psi^n)_{n \geq n_0}$ in the proof of (3.31.21), shifted down one degree, so distributivity (3.31.10).d) gives

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$$\sum_{n\geq n_0} nc_n\psi^{n-1}\psi' = \left(\sum_{n\geq n_0} nc_n\psi^{n-1}\right)\psi'$$

Reindexing $m = n-1$, $\sum_{n\geq n_0} nc_n\psi^{n-1} = \sum_{m\geq n_0-1}(m+1)c_{m+1}\psi^m = f'\circ\psi$, giving $(f\circ\psi)' = (f'\circ\psi)\cdot\psi'$. ∎

Proposition 3.31.23 (Substitution is a Ring Homomorphism)
Let A be a commutative ring and $\psi \in XA[[X]]$. Then $f \mapsto f\circ\psi$ is a ring homomorphism $A[[X]] \to A[[X]]$. If moreover A is an integral domain and the leading coefficient of ψ is a unit (so $\psi \in A((X))^\star$ by (3.31.14)) then $f \mapsto f\circ\psi$ is a ring homomorphism $A((X)) \to A((X))$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The arguments below apply to f, g, h in $A[[X]]$ in the first case and in $A((X))$ in the second.

Identity. $1\circ\psi = \psi^0 = 1$.

Additivity. Write $f = \sum_{n\geq n_0} c_nX^n$, $h = \sum_{n\geq m_0} d_nX^n$, and set $p_0 := \min(n_0, m_0)$, extending both coefficient sequences by $c_n := 0$ (resp. $d_n := 0$) for $p_0 \leq n < n_0$ (resp. $< m_0$); this changes neither series, nor the sums below, as it only adds zero terms. Then $f + h = \sum_{n\geq p_0}(c_n + d_n)X^n$, and the families $(c_n\psi^n)_{n\geq p_0}$ and $(d_n\psi^n)_{n\geq p_0}$ are summable by (3.31.21). Hence

$$\begin{aligned}(f+h)\circ\psi &= \sum_{n\geq p_0}(c_n+d_n)\psi^n = \sum_{n\geq p_0}(c_n\psi^n + d_n\psi^n) && \text{by distributivity in } A((X))\\ &= \sum_{n\geq p_0} c_n\psi^n + \sum_{n\geq p_0} d_n\psi^n = f\circ\psi + h\circ\psi && \text{by (3.31.10) } e)\end{aligned}$$

Multiplicativity. Write $f = \sum_{i\geq i_0} c_iX^i$, $g = \sum_{j\geq j_0} d_jX^j$, so $f\circ\psi = \sum_i c_i\psi^i$ and $g\circ\psi = \sum_j d_j\psi^j$, and let $P := \mathbb{Z}_{\geq i_0}\times\mathbb{Z}_{\geq j_0}$ and $a(i,j) := c_id_j\psi^{i+j}$ for $(i,j)\in P$. We have $v(a(i,j)) \geq i+j$ whenever $i+j\geq 0$ as $v(\psi^n)\geq n$ for $n\geq 0$ by the proof of (3.31.21), and for every N only the finitely many pairs with $i+j < \max(N,0)$ can fail $v(a(i,j))\geq N$, so $(a(i,j))_{(i,j)\in P}$ is summable. Then

$$\begin{aligned}(f\circ\psi)(g\circ\psi) &= \sum_{j\geq j_0}(f\circ\psi)d_j\psi^j && \text{by (3.31.10) } d)\\ &= \sum_{j\geq j_0}\sum_{i\geq i_0} c_id_j\psi^{i+j} && \text{by (3.31.10) } d)\\ &= \sum_{(i,j)\in P} a(i,j) && \text{by (3.31.11)}\end{aligned}$$

and, grouping instead by $N := i+j$, whose fibres are finite,

$$\begin{aligned}(fg)\circ\psi &= \sum_N\left(\sum_{i+j=N} c_id_j\right)\psi^N = \sum_N\sum_{i+j=N} c_id_j\psi^N && \text{by (3.3.17)}\\ &= \sum_{(i,j)\in P} a(i,j) && \text{by (3.31.10) } f) \text{ for the partition of } P \text{ by } i+j\end{aligned}$$

so $(f\circ\psi)(g\circ\psi) = (fg)\circ\psi$. ∎

3.31.1 Exponential and Logarithm

Definition 3.31.24 (Formal Exponential and Logarithm)
Let A be an integral domain with $\mathbb{Q}\subseteq A$. Define

$$\exp(X) := \sum_{n\geq 0}\frac{X^n}{n!} \in A[[X]] \qquad \log(1+X) := \sum_{n\geq 1}(-1)^{n+1}\frac{X^n}{n} \in XA[[X]]$$

For $f \in XA[[X]]$ and $g \in 1+XA[[X]]$ define $\exp(f) := \exp(X)\circ f$ and $\log(g) := \log(1+X)\circ(g-1)$, using substitution (3.31.21) (note $g-1\in XA[[X]]$).

Lemma 3.31.25 (Zero Derivative)
Let A be a commutative ring with $\mathbb{Q}\subseteq A$ and $F\in A((X))$ with $F' = 0$. Then F is a constant, that is $F\in A$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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$(F')_n = (n+1)F_{n+1} = 0$ for every $n \in \mathbb{Z}$, and $n+1$ is invertible in A for $n \neq -1$, so $F_{n+1} = 0$ for all $n \neq -1$, that is $F = F_0$. ∎

Proposition 3.31.26 (Derivatives of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$, $f \in XA[[X]]$ and $g \in 1 + XA[[X]]$. Then $\exp(f) \in 1 + XA[[X]]$ and $\log(g) \in XA[[X]]$, and

$$\exp(f)' = \exp(f)f' \qquad \log(g)' = g'g^{-1}$$

where g is a unit of $A[[X]]$ by (3.31.13).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Directly $\exp(X)' = \sum_{n\geq 1} \frac{X^{n-1}}{(n-1)!} = \exp(X)$. Since $v(f^n) \geq n$, only the term $n = 0$ contributes to the constant coefficient of $\exp(f) = \sum_n \frac{f^n}{n!}$, so $\exp(f) \in 1 + XA[[X]]$. By the chain rule (3.31.22) $\exp(f)' = (\exp' \circ f)f' = \exp(f)f'$.

log. Likewise $\log(g) \in XA[[X]]$, as each term of $\sum_{n\geq 1} (-1)^{n+1}\frac{(g-1)^n}{n}$ has order at least n. Directly $\log(1+X)' = \sum_{n\geq 1}(-1)^{n+1}X^{n-1} = \sum_{m\geq 0}(-X)^m = (1+X)^{-1}$, as $(1+X)\sum_{m\geq 0}(-X)^m = 1$ coefficientwise. By the chain rule

$$\log(g)' = (\log(1+X)' \circ (g-1))g' = ((1+X)^{-1} \circ (g-1))g' = g^{-1}g'$$

since $f \mapsto f \circ (g-1)$ is a ring homomorphism (3.31.23) and $(1+X) \circ (g-1) = g$. ∎

Proposition 3.31.27 (Functional Equations of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\exp(f+g) = \exp(f)\exp(g)$ for $f, g \in XA[[X]]$, and $\log(gh) = \log(g) + \log(h)$ for $g, h \in 1 + XA[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Write $E := \exp(f+g)$ and $U := \exp(f)\exp(g)$, a unit of $A[[X]]$ with constant coefficient 1. By the product rule and (3.31.26), $E' = E(f'+g')$ and $U' = U(f'+g')$. Hence for $R := EU^{-1}$, using the power rule (3.31.20) for U^{-1},

$$\begin{aligned} R' &= E'U^{-1} - EU'U^{-2} && \text{by the product and power rules} \\ &= E(f'+g')U^{-1} - EU(f'+g')U^{-2} = 0 \end{aligned}$$

so R is constant by (3.31.25), equal to $R(0) = 1$, that is $E = U$.

log. By the product rule and (3.31.26), $\log(gh)' = (g'h + gh')g^{-1}h^{-1} = g'g^{-1} + h'h^{-1} = (\log(g) + \log(h))'$. So $\log(gh) - \log(g) - \log(h)$ has derivative 0 and constant coefficient 0, hence is 0 by (3.31.25). ∎

Proposition 3.31.28 (exp and log are Inverse Isomorphisms)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\log(\exp(f)) = f$ for $f \in XA[[X]]$ and $\exp(\log(g)) = g$ for $g \in 1 + XA[[X]]$. Hence exp and log are mutually inverse isomorphisms of groups between $(XA[[X]], +)$ and $(1 + XA[[X]], \times)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.31.26) $\log(\exp(f))' = \exp(f)'\exp(f)^{-1} = f'$, so $\log(\exp(f)) - f$ has derivative 0 and constant coefficient 0, and is 0 by (3.31.25). For the other composite let $E := \exp(\log(g))$, so $E' = E\log(g)' = Eg'g^{-1}$ by the same result. Then

$$\begin{aligned} (Eg^{-1})' &= E'g^{-1} - Eg'g^{-2} && \text{by the product and power rules} \\ &= Eg'g^{-2} - Eg'g^{-2} = 0 \end{aligned}$$

so Eg^{-1} is constant by (3.31.25), equal to $E(0)g(0)^{-1} = 1$, that is $E = g$. Finally exp is a homomorphism by (3.31.27), with inverse log. ∎

3.31.2 Extensions of Power Series Rings

The main goal of this subsection is to show that the valuation of $k((t))$ has a **unique** extension to any finite separable extension E of $k((t))$, whose ramification index and residue degree satisfy $ef = [E : F]$ (3.31.34) and (3.31.35), and that for k algebraically closed E is itself a power series field $k((u))$ (3.31.36). The main tool is the Weierstrass division lemma for formal power series, which we record first.

Remark 3.31.29
[Che51] (Ch. II, §6) treats the case of (3.31.36) where the ramification index e is prime to $\mathrm{char}(k)$, obtaining the sharper conclusion that t itself is an e-th power in $k((u))$ (a literal Puiseux expansion $k\big(\big(t^{1/e}\big)\big)$), via Hensel's lemma applied to $W^e - v$ for the unit $v := t/u^e$. That argument is unavailable when $\mathrm{char}(k) = p \mid e$: the derivative eW^{e-1} vanishes identically, so $W^e - v$ has no simple root to lift, and t need not be an e-th power at all (an Artin-Schreier

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*extension gives a counterexample). (3.31.36) instead only produces $t = u^e c$ for **some** unit c, obtained by solving $\tau G(\tau) = u^e$ by direct coefficient-stabilisation rather than root extraction; this sidesteps the characteristic obstruction entirely, at the cost of the sharper radical form, which is in any case not needed below. The statement of (3.31.36) over an algebraically closed field, in every characteristic, is [Lan82] (Ch. I, §4, Prop. 4.3), whose proof, like ours, expands the elements of E in powers of a uniformiser u with coefficients in $k[[t]]$ and compares valuations.*

Lemma 3.31.30 (Division with Remainder by a Distinguished Power Series)
Let A be a commutative ring, $m \geq 1$ an integer and $\tau = X^m w(X)$ for $w(X) \in A[[X]]^\star$.

Then every $p \in A[[X]]$ may be written uniquely as $p = q\tau + r$ with $q \in A[[X]]$ and $r \in A[X]$ of degree $< m$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For existence, split $p = X^m p' + r$ with r the truncation below degree m and p' the shift, and put $q := p' w^{-1}$ (as $w \in A[[X]]^\star$ (3.31.13)). For uniqueness, if $q\tau + r = 0$ then $r = -q\tau \in X^m A[[X]]$ has degree $< m$, so $r = 0$; then $qX^m w = 0$ gives $q = 0$, as w is a unit and X^m is a non-zero-divisor. ∎

Lemma 3.31.31 (Free Basis for a Distinguished Power Series)
Let A be a commutative ring, $m \geq 1$ and $\tau := X^m w(X) \in A[[X]]$ with $w(X) \in A[[X]]^\star$ (and in particular $w(0) \in A^\star$ (3.31.13)).

There is a well-defined ring homomorphism $A[[Y]] \to A[[X]]$ such that $f(Y) \mapsto f(\tau(X))$, which is injective.

Furthermore $A[[X]]$ is a free $A[[Y]]$-module with basis $1, X, ..., X^{m-1}$, that is for every $p \in A[[X]]$ there are unique $f_0, ..., f_{m-1} \in A[[Y]]$ such that

$$p = \sum_{i=0}^{m-1} f_i(\tau(X)) X^i$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The homomorphism and the module structure. As $m \geq 1$ we have $\tau = X^m w \in XA[[X]]$, so $f(Y) \mapsto f(\tau)$ is a well-defined map $A[[Y]] \to A[[X]]$ (3.31.21), and it is a ring homomorphism (3.31.23) (with $\psi := \tau$). Regard $A[[X]]$ as an $A[[Y]]$-module through it, so $f \cdot p = f(\tau)p$ and $\sum_{i=0}^{m-1} f_i(\tau)X^i = \sum_{i=0}^{m-1} f_i \cdot X^i$. We must show that $1, X, ..., X^{m-1}$ span this module and are free; injectivity of the homomorphism is deduced at the end.

Spanning. Put $p_0 := p$ and, dividing repeatedly by τ (3.31.30), write $p_j = p_{j+1}\tau + r_j$ with $\deg r_j < m$. Then for every $J \geq 0$

$$p = p_{J+1}\tau^{J+1} + \sum_{j=0}^{J} r_j \tau^j, \quad \text{with } p_{J+1}\tau^{J+1} \in X^{m(J+1)}A[[X]].$$

This follows by induction on J: for $J = 0$ it is $p = p_1\tau + r_0$, and substituting $p_{J+1} = p_{J+2}\tau + r_{J+1}$ gives the step, as $\tau^{J+1} \in X^{m(J+1)}A[[X]]$ because $\tau = X^m w$. So $\left(r_j\tau^j\right)_{j\geq 0}$ is summable (3.31.9) (as $v\left(r_j\tau^j\right) \geq mj$), and $p = \sum_{j\geq 0} r_j\tau^j$ by (3.31.12), as $p - \sum_{j\leq J} r_j\tau^j = p_{J+1}\tau^{J+1} \in X^{m(J+1)}A[[X]]$ for every J. Write $r_j = \sum_{i=0}^{m-1} \gamma_{j,i} X^i$ and set $f_i := \sum_{j\geq 0} \gamma_{j,i} Y^j \in A[[Y]]$. The family $\left(\gamma_{j,i}\tau^j X^i\right)_{(j,i)}$ is summable, as $v\left(\gamma_{j,i}\tau^j X^i\right) \geq mj$. Then

$$\begin{aligned}
\sum_{i=0}^{m-1} f_i(\tau)X^i &= \sum_{i=0}^{m-1}\sum_{j\geq 0} \gamma_{j,i}\tau^j X^i \quad \text{by the definition of } f_i(\tau) \text{ and (3.31.10) } d) \\
&= \sum_{j\geq 0}\sum_{i=0}^{m-1} \gamma_{j,i}\tau^j X^i \quad \text{by (3.31.11)} \\
&= \sum_{j\geq 0} r_j\tau^j = p.
\end{aligned}$$

So $1, X, ..., X^{m-1}$ span $A[[X]]$ over $A[[Y]]$.

Freeness. Suppose that $0 = \sum_{i=0}^{m-1} f_i(\tau)X^i$ with some $f_i \neq 0$. Then because $w(0)$ is a unit we may show that $n_i := v(f_i(\tau)X^i) = mv(f_i(Y)) + i$ whenever $f_i \neq 0$. As the n_i are distinct (being so modulo m) the $\min_i(n_i)$-th coefficient is zero, a contradicton. So we conclude

$$f_0 = ... = f_{m-1} = 0.$$

Uniqueness of the f_i follows by applying freeness to the difference of two decompositions of p, and injectivity of $f \mapsto f(\tau)$ from the case $f_0 = f$, $f_1 = ... = f_{m-1} = 0$. ∎

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Example 3.31.32
Take $w = 1$, so $\tau = X^m$. For $p = \sum_{n\geq 0} a_n X^n \in A[[X]]$, writing each $n = mj + i$ with $0 \leq i < m$ gives

$$p = \sum_{i=0}^{m-1} \left(\sum_{j\geq 0} a_{mj+i} X^{mj} \right) X^i, \quad \text{so} \quad f_i(Y) = \sum_{j\geq 0} a_{mj+i} Y^j.$$

That is, f_i collects the coefficients of p whose exponent is congruent to i modulo m. For $m = 2$, f_0 collects the even-index coefficients and f_1 the odd-index ones.

Corollary 3.31.33 (Specialisation Compatibility)
In the setting of (3.31.31), let $\phi : A \to S$ be a homomorphism of commutative rings, extended coefficientwise to $\phi : A[[X]] \to S[[X]]$. Then $\phi(w) \in S[[X]]^\star$, and the decomposition of $\phi(p)$ relative to $\phi(\tau) = X^m \phi(w)$ is $\phi(p) = \sum_{i=0}^{m-1} \phi(f_i)(\phi(\tau)) X^i$, where $\phi(f_i) \in S[[Y]]$ denotes ϕ applied coefficientwise to f_i.

Proof. If $wv = 1$ then $\phi(w)\phi(v) = 1$, so $\phi(w) \in S[[X]]^\star$. Next, $\phi(f(\tau)) = \phi(f)(\phi(\tau))$ for every $f \in A[[Y]]$. Fix $N \geq 0$ and write $f = s + Y^N g$ with $s \in A[Y]$ of degree $< N$ and $g \in A[[Y]]$. Then $f(\tau) = s(\tau) + \tau^N g(\tau)$ (the tail $\sum_{n\geq N} c_n \tau^n = \tau^N \sum_n c_{n+N} \tau^n$, (3.31.9)), and likewise $\phi(f) = \phi(s) + Y^N \phi(g)$, so $\phi(f)(\phi(\tau)) = \phi(s)(\phi(\tau)) + \phi(\tau)^N \phi(g)(\phi(\tau))$. Applying the ring homomorphism ϕ to $f(\tau) = s(\tau) + \tau^N g(\tau)$ gives $\phi(f(\tau)) = \phi(s)(\phi(\tau)) + \phi(\tau)^N \phi(g(\tau))$, so $\phi(f(\tau)) - \phi(f)(\phi(\tau)) \in \phi(\tau)^N S[[X]] \subseteq X^{mN} S[[X]]$, as $\phi(\tau) = X^m \phi(w)$. This holds for every N and $\bigcap_N X^N S[[X]] = 0$ (3.31.6), so $\phi(f(\tau)) = \phi(f)(\phi(\tau))$. Applying ϕ to $p = \sum_i f_i(\tau) X^i$ therefore gives $\phi(p) = \sum_i \phi(f_i)(\phi(\tau)) X^i$. This exhibits $\phi(p)$ in the form guaranteed unique by (3.31.31) over S, so it is **the** decomposition of $\phi(p)$. ∎

Lemma 3.31.34 (Local Degree at a Prime above (t))
Let k be a field, $F := k((t))$, E/F a finite separable extension of degree n, C the integral closure of $k[[t]]$ in E, and $\mathfrak{p} \subset C$ a prime with $\mathfrak{p} \cap k[[t]] = (t)$ (such a prime exists). Then $B := C_{\mathfrak{p}}$ is a DVR with fraction field E and $k' := C/\mathfrak{p}$ is a finite field extension of k. Write $e := v_B(t)$ and $f := [k' : k]$. Then $ef = n$. More precisely, if u is a uniformiser of B and $\omega_1, \ldots, \omega_f \in C$ map to a k-basis of k', then the elements $\omega_j u^i$ $(0 \leq i < e,\ 1 \leq j \leq f)$ form an F-basis of E, and $B = \bigoplus_{i,j} k[[t]] \omega_j u^i$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Step 1: a DVR. Let $\mathcal{O}_F := k[[t]]$, a DVR with fraction field F (3.31.16) (so a Dedekind domain (3.30.3)). By (3.22.28) C is a finite $\mathcal{O}_F$-module, and by (3.30.11) it is Dedekind. By lying-over (3.22.18) there is a prime $\mathfrak{p} \subset C$ with $\mathfrak{p} \cap \mathcal{O}_F = (t)$. Then $B := C_{\mathfrak{p}}$ is a DVR with fraction field E, and $e := v_B(t) \geq 1$.

Step 2: residue field and valuation. $C/\mathfrak{p}$ is a domain, finite over $\mathcal{O}_F/(t) = k$, so $k \hookrightarrow C/\mathfrak{p}$ is injective and integral, hence $C/\mathfrak{p}$ is a field (3.22.14), finite over k. As $B = C_{\mathfrak{p}}$, (3.5.43) gives $B/\mathfrak{m}_B \cong \mathrm{Frac}(C/\mathfrak{p}) = C/\mathfrak{p} = k'$. Every $a \in \mathcal{O}_F^\times$ is a unit of B, so $v_B(x) = e v_F(x)$ for $x \in F^\times$. Let u be a uniformiser of B, so $t = u^e w$ with $w \in B^\star$, and let $\omega_1, \ldots, \omega_f \in C$ map to a k-basis of k'.

Step 3: spanning. Every $b \in B$ can be written $b = \sum_{i<e} \sum_j a_{ij} \omega_j u^i$ with $a_{ij} \in k[[t]]$. We show by induction on $N \geq 0$ that

$$b \equiv \sum_{i<e} \sum_j \left(\sum_{l<N} c_{ijl} t^l \right) \omega_j u^i \pmod{t^N B}$$

for some $c_{ijl} \in k$. The case $N = 0$ is trivial. For the inductive step, write the current remainder as $t^N y$ with $y \in B$. Since $B/uB = k'$ has k-basis the images of the ω_j, repeated division by u gives $y = \sum_{i<e} \sum_j d_{ij} \omega_j u^i + u^e y'$ with $d_{ij} \in k$, $y' \in B$. Using $u^e = tw^{-1}$,

$$t^N y = \sum_{i<e} \sum_j (t^N d_{ij}) \omega_j u^i + t^{N+1} (w^{-1} y'),$$

with $w^{-1} y' \in B$; so taking $c_{ijN} := d_{ij}$ advances the induction, with new remainder in $t^{N+1} B$. Put $a_{ij} := \sum_l c_{ijl} t^l \in k[[t]] \subseteq C \subseteq B$. Then $z := b - \sum_{i,j} a_{ij} \omega_j u^i$ satisfies $z \in t^N B$ for every N, so $v_B(z) \geq eN$ for all N and $z = 0$.

Step 4: independence, basis and degree. Every element of E has the form $t^{-N} b$ with $b \in B$, so Step 3 gives $E = \sum_{i<e,,j} F \omega_j u^i$. Let $a_{ij} \in F$ be not all zero. For each i with some $a_{ij} \neq 0$ put $m_i := \min_j v_F(a_{ij})$ and write $a_{ij} = t^{m_i} a'_{ij}$ with $a'_{ij} \in k[[t]]$ and $a'_{ij}(0) \neq 0$ for some j. As $a'_{ij} \equiv a'_{ij}(0) \pmod{tk[[t]]}$ and $t \in uB$,

$$\sum_j a'_{ij} \omega_j \equiv \sum_j a'_{ij}(0) \omega_j \pmod{uB},$$

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which is non-zero in k' since the images of the ω_j are k-linearly independent. So $\sum_j a'_{ij}\omega_j$ is a unit of B and $v_B\left(\sum_j a_{ij}\omega_j u^i\right) = em_i + i$. These valuations are distinct modulo e for distinct i, so the minimum over i is attained exactly once and $\sum_{i,j} a_{ij}\omega_j u^i \neq 0$ (3.23.2). Hence the $\omega_j u^i$ are linearly independent over F, so form an F-basis of E and $n = ef$.

Moreover for $b = \sum_{i,j} a_{ij}\omega_j u^i$ with $a_{ij} \in F$ the same computation gives $v_B(b) = \min_i(em_i + i)$. If $b \in B$ then $em_i + i \geq 0$ with $0 \leq i < e$, so $m_i \geq 0$ and all $a_{ij} \in k[[t]]$. Together with Step 3 this gives $B = \bigoplus_{i,j} k[[t]]\omega_j u^i$. ∎

Corollary 3.31.35 (Uniqueness of the Extended Valuation)
Let k be a field, $F := k((t))$, E/F a finite separable extension of degree n and C the integral closure of $k[[t]]$ in E. Then C has exactly one prime $\mathfrak{p}$ lying over (t), and its ramification index e and residue degree f satisfy $ef = n$. Moreover the DVR $C_\mathfrak{p}$ is the unique valuation ring of E containing $k[[t]]$ whose maximal ideal contracts to (t). If k is algebraically closed then $f = 1$ and $e = n$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\mathfrak{q}$ be any prime of C over (t), which exists by lying-over. By (3.31.34) its ramification index and residue degree satisfy $e_\mathfrak{q} f_\mathfrak{q} = n$.

Now apply the Ramification Formula (3.30.20) to the AKLB situation $k[[t]] \subset C$ over E/F (both Dedekind by (3.30.11), and E/F separable), at the prime (t):

$$\sum_{\mathfrak{q}\mid(t)} e_\mathfrak{q} f_\mathfrak{q} = [C/tC : k] = n.$$

Each term equals n by the first paragraph, so there is exactly one prime $\mathfrak{p}$ over (t) and $e_\mathfrak{p} f_\mathfrak{p} = n$. If k is algebraically closed then $C/\mathfrak{p}$, a finite extension of k, equals k, so $f_\mathfrak{p} = 1$ and $e_\mathfrak{p} = n$.

For uniqueness of the valuation ring, let $V \subseteq E$ be a valuation ring with $k[[t]] \subseteq V$ and $\mathfrak{m}_V \cap k[[t]] = (t)$. By (3.23.14), $C \subseteq V$. Then $\mathfrak{m}_V \cap C$ is a prime of C over (t), hence equals $\mathfrak{p}$. Every element of $C \setminus \mathfrak{p}$ lies outside $\mathfrak{m}_V$, so is a unit of V; hence $C_\mathfrak{p} \subseteq V$. Since $C_\mathfrak{p}$ is a valuation ring of E and valuation rings are maximal for domination among proper subrings of E (3.23.2), $V = C_\mathfrak{p}$. ∎

Lemma 3.31.36 (Finite Separable Extensions of $k((t))$ are Power Series Fields)
Let k be algebraically closed, $F := k((t))$, and E/F a finite separable extension of degree n. Then there are $u \in E$, an element $\tau \in uk[[u]]$ with $\tau = u^n c$ for some $c \in k[[u]]^\star$, and a field isomorphism $\Psi : E \xrightarrow{\sim} k((u))$ with $\Psi|_F$ the substitution $t \mapsto \tau$ and $\Psi(u) = u$. In particular $1, u, \ldots, u^{n-1}$ is an F-basis of E.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Step 1: reduction to the local degree. Let C be the integral closure of $k[[t]]$ in E and $\mathfrak{p} \subset C$ a prime over (t), which exists by lying-over, and put $B := C_\mathfrak{p}$. Its residue field $C/\mathfrak{p}$ is finite over the algebraically closed field k (3.31.34), so equals k and $f = 1$. Take $\omega_1 := 1$. Then (3.31.34) gives $e = n$, that $1, u, \ldots, u^{n-1}$ is an F-basis of E for a uniformiser u of B, and $B = \bigoplus_{i<n} k[[t]]u^i$. Write $t = u^n w$ with $w \in B^\star$.

Step 2: the minimal polynomial. Write $w^{-1} = \sum_{i<n} g_i u^i$ with $g_i \in k[[t]]$ (Step 1). The constant $g_0(0)$ is the residue of w^{-1}, so $g_0(0) \in k^\star$. From $u^n = tw^{-1}$, u is a root of

$$P(Y) := Y^n - t\sum_{i<n} g_i(t)Y^i \in F[Y].$$

Since $F[u] = E$ has degree n over F, P is the minimal polynomial of u, and $E \cong F[Y]/(P)$ with $Y \mapsto u$.

Step 3: solving for τ. For $s \in uk[[u]]$ put $G(s) := \sum_{i<n} g_i(s)u^i \in k[[u]]$, where $g_i(s)$ is the substitution (well defined as s has zero constant term). Since $G(s) \equiv g_0(0) \pmod{u}$, $G(s)$ is a unit. Define $T(s) := u^n G(s)^{-1} \in u^n k[[u]]$. For $s \equiv s' \pmod{u^N}$ we have $g_i(s) - g_i(s') \in (s - s')k[[u]] \subseteq u^N k[[u]]$, hence $G(s) \equiv G(s') \pmod{u^N}$ and $T(s) \equiv T(s') \pmod{u^{N+n}}$. Starting from $s_0 := 0$ and setting $s_{m+1} := T(s_m)$, the sequence satisfies $s_{m+1} \equiv s_m \pmod{u^{m+n}}$, so each coefficient stabilises. Let $\tau \in uk[[u]]$ be the coefficientwise limit. Then $\tau \equiv T(\tau) \pmod{u^N}$ for every N, so $\tau = T(\tau)$, i.e. $\tau G(\tau) = u^n$. Also $\tau = u^n c$ with $c \in k[[u]]^\star$, since $\tau G(\tau) = u^n$ and $G(\tau)$ is a unit.

Step 4: the homomorphism. The substitution $t \mapsto \tau$ defines a ring homomorphism $\sigma : k[[t]] \to k[[u]]$ (3.31.23) (as τ has zero constant term), and since τ is a unit times u^n it extends to a field homomorphism $\sigma : F \to k((u))$. The homomorphism $F[Y] \to k((u))$, $Y \mapsto u$, extending σ, kills P:

$$\sigma(P)(u) = u^n - \tau\sum_{i<n} g_i(\tau)u^i = u^n - \tau G(\tau) = 0.$$

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So it factors through $E \cong F[Y]/(P)$, giving a field homomorphism $\Psi : E \to k((u))$ with $\Psi|_F = \sigma$ and $\Psi(u) = u$. It is injective, as E is a field.

Step 5: surjectivity. Since $\tau = u^n c$ with $c \in k[[u]]^\star$, the Weierstrass division lemma (3.31.31) (with $A := k$, $X := u$, $m := n$, $w := c$, so the series of the lemma is τ) gives $k[[u]] = \bigoplus_{i<n} \sigma(k[[t]])u^i$. Multiplying by powers of τ gives $k((u)) = \sum_{i<n} \sigma(F)u^i = \sigma(F)[u]$. But $\operatorname{im}\Psi = \Psi(F[u]) = \sigma(F)[u]$, so Ψ is surjective.

Hence $\Psi : E \xrightarrow{\sim} k((u))$ is an isomorphism of fields over F, with $\Psi(u) = u$ and $\Psi(t) = \tau = u^n c$. ■

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